

# Understanding Transportation as System-of-Systems Design Problem

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The purposes of this paper are to introduce an emerging class of problems called system-of-systems, present the primary traits of the class, and then described the relevant implications for the aerospace design community. Two primary traits, evolutionary and emergent behavior, are highlighted since they call attention to a network-of-systems, dynamic-behavior focus as opposed to an individual system, static-behavior focus. Further, methodological needs for investigation of these traits lie outside present capability in the community. To explore these implications in depth, the casting of future transportation concepts as a system-of-systems is described. A recently developed “proto-method” for this class of problems is presented and explored in the context of the transportation domain, including promising approaches for building effective modeling and simulation. The casting of civil transportation as a system-of-systems problem is important as much of the literature on system-of-systems (limited as it is) resides primarily in the defense systems domain.

## I. Introduction

PROBLEMS of significant complexity are facing decision-makers within government and industry, and the moniker of “system-of-systems” (heretofore shortened to SoS) is increasingly being applied to problems of this type. Effective system analysis for decision-support quickly becomes unmanageable in this context, with multiple, heterogeneous, distributed systems involved and couched in networks at multiple levels. The situation is exacerbated by the fact that most organizations (and their processes) remain in the “stovepipe” mode- their people and tools are configured to study within narrow bins with little or no analysis across those bins. Current frames of reference, thought processes, analysis, and design methods are not complete for these SoS problems, exemplified in this paper by future national transportation system realizations. A holistic framework is needed that enables decision makers to discern whether related infrastructure, policy, and/or technology considerations together are good, bad (or indifferent) over time.<sup>1</sup> And this need is urgent, for the SoS type almost always involves decisions that commit large amounts of money, for which ultimate failure or success carries heavy consequences over several generations, and that impact large segments of the public. The costs associated with a chosen path may still be high, but the intent is to maximize the probability that these costs result in good consequences rather than bad.

A small number of discernable groups of researchers (themselves small in size) have begun formal study of this problem class from distinct points of origin, thus bringing different perspectives, lexicons, and sets of preferred techniques. However, it is unlikely that any single set of algorithms from traditional academic domains can solve the SoS problem. Thus, in addition to new methods, *integration interfaces* are needed at multiple levels to understand the problem in full context and ensure that interconnections between related algorithms can be exploited. Departing from the traditional practice of individual system optimization, a tenet of both SoS methods and solutions must be the recognition that the organization of systems/algorithms is just as important as the nature of the systems/algorithms to be organized. This philosophical connection between traits of the problem and traits of the method can be exploited when planning implementation via object-oriented approaches. This is explored further later in the paper.

Thus, this particular research endeavor has significance both from an intellectual and practical perspective. For the former, it is important to begin to establish a solid foundation for recognizing patterns and structure in SoS problems; for the latter, there is hope for development of improved tools for decision-makers grappling with difficult problems, such as the transformation of the transportation enterprise.<sup>†</sup>

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† Although transportation is the application topic of this paper, the nation’s health care and energy systems are additional, prime examples of such vexing challenges.

## A. Current Views

A symposium on system-of-systems was recently held seeking common understanding and identification of the most pressing open questions. The participants constituted a diverse group of individuals from government, academia, and industry struggling to understand SoS. The gathering's conversation focused on the status of the thinking and current practices (ad-hoc or otherwise) associated with SoS.. A report of the salient outcomes from the symposium is available in Ref. [2]. Some of the consensus findings are summarized here. First, system-of-systems was deemed a new problem class (or at least a much larger and more vexing form of existing problems), though not yet well-formed into a definable field of study. Second, existing fields of study provide incomplete models and partial solutions but are important for the intellectual foundation for SoS study. Finally, there is an information gap between the data produced by engineers/analysts and the insights best utilized by decision-makers. This gap can result in much effort being expended for little benefit (to decision-maker and society both). Worse, if the decision space can be likened to a minefield, poor understanding can result in the triggering of a mine in the minefield. There were other findings highlighting the myriad of disciplines likely to be needed, prompting some to call for "a system-of-systems study of system-of-systems research", certainly a prodigious self-referential study if there ever was one.

The first finding deals in part with the age-old question: What's in a name? While names are the compact, convenient labels we use to point to much larger and more complex concepts, they can also be a distraction. So it is with system-of-systems. The precise meaning of this three-word-hyphenated label is not nearly as important as the problem type that it (imperfectly) represents; thus, we will focus on the nature and distinguishing traits of this problem class rather than attempting to forge a definitive meaning.

The second finding, related in some sense to the first, concerns the state of affairs among those that are seeking to understand system-of-systems. The application domain in which "system-of-systems" most often appears is that of military systems development and acquisition. It is now part and parcel of major programs, appearing throughout key documents such as the Operational Requirements Document (ORD) and forcing a reexamination of especially systems engineering. Emerging from the network centric warfare concept<sup>3</sup>, system-of-systems is part of the strategy by which independent but related platforms are to be procured, fielded, and manipulated according to warfighter needs in any particular circumstance. Thus, the concept of connectivity between heterogeneous systems comes to the fore in all stages of systems design and development, including very early phases where joint warfighter requirements are being discerned at the same time as technologies are being explored.<sup>4</sup> However, even the strongest advocates of the SoS perspective would admit that the methods are not yet complete in this domain; further, it remains unclear whether they transmute appropriately to other applications, especially those in the commercial sector. This latter observation motivates the present research.

As a result of defense system related transformations, perhaps the most active group discussing SoS is the systems engineering community. Several contributions in the recent literature have come from this community where there is a growing recognition that systems engineering processes are not complete for the new challenges posed by SoS.<sup>5</sup> Specifically, Sage presented a working collection of traits for SoS that points to a "new federalism" as a construct for dealing with SoS.<sup>6</sup> In this setting, the federalism offers a means of understanding the trait of operational independence. Earlier, the work of Maier was important in delineating the salient traits of SoS that will be explored later in this paper.<sup>7</sup>

Operations Research (OR) is another large body of thought for which SoS has kinship. The diversity of OR results in several of its thrusts (transportation logistics, network design, etc.) having relevance for SoS. Even before the primary developments in OR, the idea of General System Theory (later often known simply as system theory) which traces back to Ludwig Bertalanffy's *General Systems Theory* certainly also rings true to the SoS challenge.<sup>8</sup> The subject matter of this theory is the formulation of principles that are valid for any "system" whatever the nature of components. Similarities of observed phenomena across fields of study lend credence to the idea that commonalities can be instructive and useful. The prime thrust is a common set of principles applicable to any system across scientific fields. The focus on interconnections within systems, in general, is an important aspect for SoS, and especially important for the "unintended consequences" behavior that will be mentioned later. Modern proponents of this line of thinking include the very well-known work of Ackoff, who has proposed that the real roots of OR are/should be in General Systems Theory.<sup>9</sup>

Understanding the implications of interconnections leads to another community of interest- network theory. The network theory community examines basic structure and behavior in networks that appear in the seemingly vastly different domains- biology, computer science, and sociology, etc. The work of Barabasi<sup>10</sup> is an example of a journey from the mathematical structure of networks to the fascinating correspondence with actual data from networks as diverse as the disease transmission, the World Wide Web, and air transportation. These networks exhibit so-called "small world" behavior, fundamentally different than the predicted from standard random networks, and such

behavior has large implications for robustness of real networks of systems. Already one recent study has been published to begin to map these concepts more rigorously to air transportation applications.<sup>11</sup>

The final community of interest to be discussed in relation to SoS research is that of aerospace design (of which the author counts himself as member). A primary motivation for this paper is to prompt further examination of the distinctions between SoS problems and traditional problems in aerospace design. To begin the discussion, then, a brief review of the key methodological advancements in aerospace systems design is in order. First, aerospace design has most often focused on singular product (system) design. Design method innovations have come primarily in how the system is characterized (i.e. treatment of the constituent disciplines and design variable vector, ( $\mathbf{x}$ ) and what are the objectives ( $\mathbf{y}$ ) used to compare alternative solutions and/or search for the best (i.e. optimization). This evolution is summarized in very broad themes in Table 1. The Design-for-Performance philosophy seeks the singular solution that maximized performance needs of the customer. The arrival of emphasis on affordability (performance per unit cost) changed matters. For example, the advent of concurrent engineering (adding especially a process perspective) and then multidisciplinary design optimization, MDO, (adding especially formal treatment of the interacting disciplines) gave rise to new methods that led to better solutions through exploitation of knowledge of disciplinary interactions. Subsequently, advancement in parametric methods applied to aerospace design improved matters further by allowing higher fidelity contributing analyses to be employed and more clearly exposing these interaction sensitivities in the resultant multidimensional design space. Finally, realization of the importance of uncertainty led to the robust design family of approaches. Attributions for all these innovations are far too numerous to cite here and their details are not germane to this paper, but accessing good survey articles would be recommended for further exploration of these topics.

**Table 1. Brief Synopsis of Aerospace Design Methods Innovation**

| Philosophy               | Method                                                               | Synopsis                                                                                                      |
|--------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Design for Performance   | Deterministic Optimization                                           | $\max y(\mathbf{x})$                                                                                          |
| Design for Affordability | Concurrent Engineering & Multidisciplinary Design Optimization (MDO) | $\max y(\mathbf{x}_{aero}, \mathbf{x}_{struct}, \mathbf{x}_{coupling})$                                       |
|                          | Parametric Design (metamodels)                                       | $y = b_o + \sum_{i=1}^k b_i x_i + \sum_{i=1}^k b_{ii} x_i^2 + \sum_{i=1}^{k-1} \sum_{j=i+1}^k b_{ij} x_i x_j$ |
|                          | Robust Design                                                        | $\max P(y(\mathbf{x}) > y_{nom})$                                                                             |

Implicit in all the aerospace design methods is some sense of self-contained purpose for the vehicle as specified, for example, by the design mission. One of the questions being proposed in this paper is: What is next set of innovations that improve the effectiveness of methodology? An understanding of system-of-systems for which air vehicles are a primary part may be the answer. If an air vehicle is a mere part of a SoS, then both the understanding of objectives  $\mathbf{y}$  and relevant subset of  $\mathbf{x}$  depends on this larger context. The advent of proposed revolutionary air vehicle concepts (e.g., uninhabited air vehicles, personal air vehicles) and next-generation airspace networks as part of a revamped transportation system are indicative of this scenario. What then is the overall vehicle design problem in this context? What are the new objectives and contributing disciplines that must be brought to bear? In the answers to these questions, the innovations in aerospace systems design do point the way. As MDO sought optimality through couplings between disciplines (with integrated objective functions), SoS methods must explore trans-domain interactions and cooperation among independent systems. A challenge will be that this may demand *sub-optimality* along some axes of the vehicle objective space for the betterment of the entire SoS. Additionally, as robust design sought to find configurations that performed well in the face of modeled uncertainty, SoS solutions must be found that are robust (i.e. asymptotically good over time) in the face of an ensemble of plausible futures.<sup>12</sup> But, as seen shortly, there are SoS traits that require advancements beyond the insights & tools available to us today.

## B. Purposes of this Paper

Indeed, the diversity of those groups in the vanguard of system-of-systems indicates both the promise and challenges that lie ahead. It is quite likely that most in these groups have an intuitive sense of the nature of this new problem class- especially that they are “messy” and that they generate the “unintended consequences” that so often, arise and usually accompanied by dramatic reductions in efficiency and performance. But concrete methods remain

elusive because the problem itself is ill-defined. A mission statement for the on-going research could be summarized as this: generation of system-of-systems formulations/tools/processes to understand and bound complex problems and create an ability to determine sensitivities and guide policies/decisions/visions.

The purpose of this paper is then threefold. First, the distinguishing traits of the system-of-systems problem class (as currently understood) are presented. Second, initial strategies for understanding, characterizing, and measuring the goodness of them are discussed. Finally, initial application of strategies to the study of future transportation concepts and implications for modeling and simulation are addressed.

## II. General Problem Characterization

### A. Distinguishing Traits of SoS

As addressed in the introduction, attempts to categorically define the system-of-systems problem are counterproductive, since it focuses on artifact (the name) as opposed to the challenge itself. Though we may not wish to precisely define SoS, it is important to understand and recognize the traits of the problem type. We wish to know one if we see one; further, the development of effective methods will depend on this understanding. An introductory description of the traits of a SoS problem is offered in the section along with a proposed lexicon to communicate with others about them. The set of traits are presented in Table 2, including a description of the trait and comments on the difficulties that arise for aerospace design methods. When a preponderance of the traits is observed<sup>6</sup>, one can be fairly certain that a SoS problem is at hand and that characteristic behaviors can be expected.

Two general observations are also relevant. First, the traits as described provide a rationale for why “system-of-systems” is not simply a collection of systems. There is a richer meaning than this sort of simplistic ‘box-inside-a-box’ approach. Second, some of the traits are more crucial to truly delineating why a SoS is a unique design problem. One of these, certainly, is the emergence property. If a collection of systems displays no emergent properties, than it would be possible to completely analyze the collection by adding the sum of its individual parts. Other traits, such as the evolutionary nature, are certainly important for the analysis complexity they introduce, but, for example, many dynamic systems can be considered to change internal structure during operation.

**Table 2. Distinguishing Traits of System-of-Systems**

| Trait                                 | Description                                                                                                                                                    | Comment                                                                                                                                                                                                                                                                                       |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operational & Managerial Independence | Constituent systems are useful in their own right and generally operate independent of other systems (i.e. with unique intent provided by the owner/operator). | <ul style="list-style-type: none"> <li>- Discerning the systems coupling variables is not easy.</li> <li>- At the SoS level, <i>no one is completely in charge</i>.</li> <li>- As described in Ref. [6], a SoS will thus most likely represent a federation, perhaps formed ad-hoc</li> </ul> |
| Geographic Distribution               | Constituent systems are not physically co-located; but, they be in communication.                                                                              | <ul style="list-style-type: none"> <li>- Spatial dynamics important for SoS behavior</li> <li>- Communication performance (information sharing)</li> </ul>                                                                                                                                    |
| Evolutionary Behavior                 | The SoS is never completely, finally formed, constantly, changes and has a “porous” problem boundary; i.e. is a living system.                                 | <ul style="list-style-type: none"> <li>- Behavior is time varying, not well-treated in aerospace design methods to date</li> <li>- Time epochs span “Generational time”</li> </ul>                                                                                                            |
| Emergent Behavior                     | Properties appear in the SoS that are not apparent (or predicted) from the constituent systems.                                                                | <ul style="list-style-type: none"> <li>- Results in non-monotonic behavior</li> <li>- Not readily addressed in current design methods</li> <li>- Its “amount” depends on knowledge of SoS levels.</li> </ul>                                                                                  |
| Networks*                             | Networks define the connectivity between independent systems in the SoS through rules of interaction.                                                          | <ul style="list-style-type: none"> <li>- Existing design methods rarely deal with networks</li> <li>- Topology and adaptability of the networks are crucial to SoS performance and are the source of emergence</li> </ul>                                                                     |
| Heterogeneity*                        | Constituent systems are of significantly different nature, with different elementary dynamics that operate on different time scales.                           | <ul style="list-style-type: none"> <li>- Primarily a challenge in modeling complexity</li> <li>- Existing methods focus on single system design</li> <li>- Variety of system analysis techniques will be required</li> </ul>                                                                  |
| Trans-domain*                         | (Proposition) Effective study of SoS requires unifying knowledge across fields of study: engineering $\cup$ economy $\cup$ policy $\cup$ operations            | <ul style="list-style-type: none"> <li>- Dramatic advancement in problem definition, lexicon and domain interfaces are needed (the spirit of MDO, but across domains instead of disciplines).</li> </ul>                                                                                      |

\* These traits identified as important by the author above and beyond the traits listed in Ref. [6] and sources cited therein.

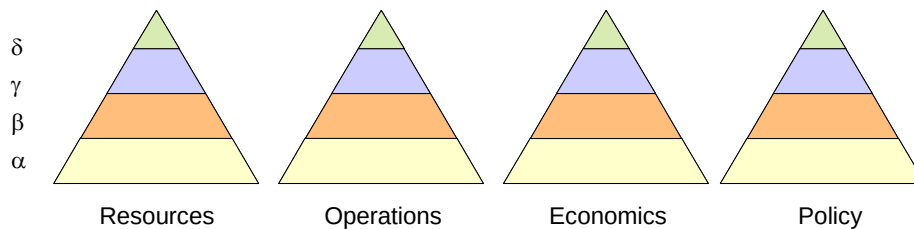
## B. Problem Formulation for SoS- An Initial Approach

The traits summarized above are useful in sketching the dimensions of the SoS problem and collection of the challenges is important because they will have to be overcome eventually. Both the traits and the challenges point towards key structures whose relevance will be clear when examining simulation approaches. However, a prerequisite ingredient is an effective language to allow communication among diverse parties over this trans-domain landscape. In this section of the paper, steps that have been taken towards generating proto-methods for analyzing SoS problems, beginning with a useful language, are presented.

This “embryonic” lexicon framework consists of two major structures (Table 3)<sup>1</sup>: categories of systems and levels of organization. Four levels and four categories are shown; however, there can be any number of levels and categories depending on the circumstance. For each category, there is a hierarchy of levels. To avoid the confusion with ambiguous derivations (e.g. system  $\rightarrow$  system-of-systems  $\rightarrow$  architecture?!), the lexicon employs the unambiguous use of Greek letters to establish the hierarchy. Alpha ( $\alpha$ ), Beta ( $\beta$ ), Gamma ( $\gamma$ ), and Delta ( $\delta$ ) indicate the relative position within each category. The collection of  $\alpha$  entities and their connectivity determines the construct of a  $\beta$ -level network; likewise, a  $\gamma$ -level network is an organized set of  $\beta$  networks. Hence, the  $\delta$ -level can be described as a network with varying levels of  $\alpha$ ,  $\beta$ ,  $\gamma$  networks. The graphic in Figure 1 unfolds a pyramid that represents a first attempt at expressing the organizational idea behind the foundation for a SoS graphical representation model. Later use of this concept in the transportation example will show that the relative “width” of each section (a measure of number of entities at that level) is not properly captured by the pyramid- there likely will be orders of magnitude difference between  $\alpha$ -level and  $\beta$ -level.

**Table 3. An Example System-of-Systems Lexicon: Categories & Levels**

| Categories         | Descriptions                                                                                                                                      |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Resources          | The entities (systems) that give physical manifestation to the system-of-systems                                                                  |
| Economics          | The non-physical entities (stakeholders) that give intent to the SoS operation                                                                    |
| Operations         | The application of intent to direct the activity of physical & non-physical entities                                                              |
| Policies           | The external forcing functions that impact the operation of physical & non-physical entities                                                      |
| Levels             | Descriptions                                                                                                                                      |
| Alpha ( $\alpha$ ) | The base level of entities, for which further decomposition will not take place. $\alpha$ -level components can be thought of as building blocks. |
| Beta ( $\beta$ )   | Collections of $\alpha$ -level systems, organized in a network.                                                                                   |
| Gamma ( $\gamma$ ) | Collections of $\beta$ -level systems organized in a network.                                                                                     |
| Delta ( $\delta$ ) | Collections of $\gamma$ -level systems organized in a network.                                                                                    |



**Figure 1. Categories and Hierarchy: An Unfolded Pyramid Unifying the Lexicon**

This framework intuitively points out an important point about SoS problems: *the behavior of the SoS is dominated by the structure and organization at higher levels as opposed to the characteristics of the  $\alpha$ -level entities.* This also resonates with the importance ascribed to the emergence property which, by definition, can only be perceived at the  $\beta$ -level or above. In other words, from a design perspective, a relevant question might be: How does the preferred or observed behavior at the upper levels (e.g.  $\gamma$ -level) affect the possibilities for alternatives at the lower levels ( $\alpha$  and  $\beta$ )? Not coincidentally, these upper levels are also the levels where the most consequential decisions arise.

### C. Evolutionary and Emergent Traits- Implications for SoS Modeling

The conceptual framework and lexicon already indicates the modeling and measuring performance in a SoS context will require new approaches. In this section, the current approach in aerospace systems design is reviewed and then contrasted with SoS needs. In particular, the implications that arise from the evolutionary nature (calling for a dynamic response measure) and emergence (introducing non-monotonic behavior) are discussed.

#### 1. State-of-practice: robust design

The robust design method for aerospace systems was introduced only briefly in Table 1. Though there are a variety of formulations reported in the literature, one popular one combines sampling-based and simulation-based perspectives. Typically, a Monte Carlo sampling technique operates on expected variability in selected parameters in the simulation model expressed via probability distribution functions. The simulation model (or the surrogates models, or metamodels) that describe the system remain fixed. This method produces outcome probability distributions that estimate likely performance.<sup>13</sup> This and related approaches<sup>14</sup> are important for moving beyond the deterministic design paradigm and for providing meaningful, comprehensible outcomes for most decision-makers. Yet, still only a point estimate is produced, fixed in time and tied to the uncertainty distributions chosen. The latter observation illustrates why these are primarily *risk-based approaches*, rather than uncertainty-based, since the variability of uncertain parameters is assumed known.

#### 2. Evolutionary Nature

In contrast, for a SoS problem, both the unknown variations in external factors and the internal structure of the of the SoS may change with time. The evolutionary trait of SoS can be traced to two important points- most SoS applications of interest already exist and their *internal structure* does and will change over time. The National Transportation System (NTS), to be discussed, is a clear example. The NTS operates today and will do so even after it is transformed; however, at future points in time its constituent systems and networks will be different. Legacy systems will mix with new systems and operational paradigms. Evolutionary behavior implies that the genetics of what exists today must be understood and considered when analyzing future realizations.

All dynamic systems exhibit evolutionary behavior in some sense. Studies in control theory have developed many methods to deal with them. There are unique aspects of SoS problems that complicate the measurement of robustness over time. The following example illustrates the insight, captured in Figure 2, and highlights that both heterogeneity and networks traits are at play. Initially, three heterogeneous systems at the  $\alpha$ -level of a SoS are presumed present at time  $t_1$ , configured in a particular  $\beta$ -level network,  $\beta_1$ . Other  $\beta$ -level networks may certainly exist (not shown), themselves organized at the  $\gamma$ -level. As time evolves, the constituent systems, their connectivity to others, and their self-interested objectives may change. Over a long enough period, a significant portion of the initial set of  $\alpha$ -units present at  $t_1$  could have been replaced. Yet, from the design perspective, we still speak of the same entity, the “so-and-so system-of-systems”, as the subject of our design. As the evolution depicted Figure 2 shows, two new  $\alpha$ -level systems appear and by the third time epoch the connectivity of the SoS is significantly changed. It would be expected that the SoS behavior/performance will certainly evolve along with these internal changes. This behavior is not readily captured by current approaches in robust design methods.

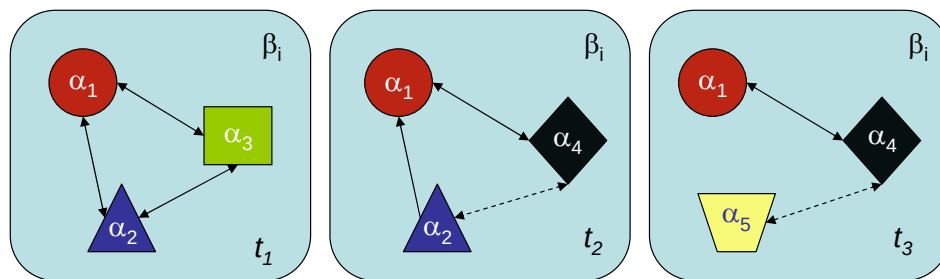


Figure 2. The evolving nature of the SoS: shifting constituents and connectivity

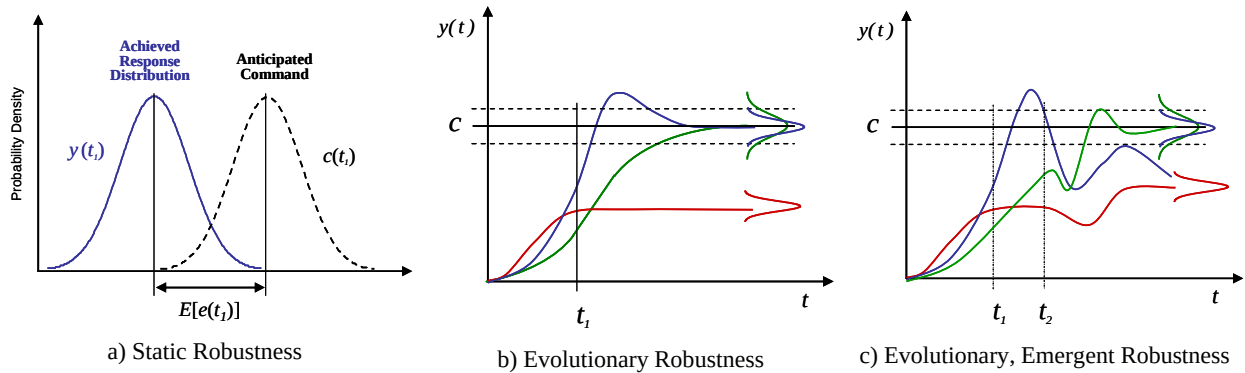
#### 3. Emergent Properties

The concept of emergence has now been mentioned twice. First, its importance amongst SoS traits was proposed when the collection of traits was introduced. Second, the appearance of emergent properties occurs (and is detected) only at the upper levels of the SoS hierarchy. Both statements are consistent with the widely accepted, if informal, definition that emergence implies that the “whole is greater than the sum of the parts”. However, acknowledging the emergence may occur, while interesting, does not immediately assist in forming mental (and, later, simulation) models of SoS problems and using these models to make better design decisions. It would of value to be able to

recognize recurring patterns of emergence and thus seek to identify underlying mechanisms at play.<sup>15</sup> To do this, then, understanding the impact of emergence on the time-dependent behavior that results from the evolutionary behavior is required. Further, the proper analysis methods for ensuring the emergent properties can be examined must be determined; on this issue, it is proposed that simulation is the optimum method for this task. This latter topic will be covered later in this paper.

#### 4. Interim Summary

The fact of evolving structure of SoS problems presents real challenges for performance measurement for now we must deal with in time-varying (and possibly non-monotonic systems). SoS performance may depend not only on what is being measured but also over what time epoch and under what bounds on uncertainty. The implications of these considerations are summarized in Figure 3. In part a), the static, robust design approach compares achievable and required (commanded) performance distributions at a given time  $t_i$ . An error signal results when there is positive probability that achieved may not satisfy required- the expected value of this error signal,  $E(t_i)$ , measures the robustness. In part b), SoS performance against a step command signal (requirement) is shown as a notional second order time response. Three candidate solutions are shown and a whole new set of performance measures arise. How quickly does each respond (e.g. rise time)? Is the response smooth (e.g. critically damped)? Does the response meet the requirement in given time? (e.g. steady-state error). It is also important not to be deceived by the “normal” looking time axis in this chart ... for SoS, these time responses may be *generational in scale*. Finally, in part c), the result of emergent behavior is viewed as the onset of non-monotonic response, indicating that a SoS behavior, when measured at  $\beta$ -levels and above, does not settle before the emergent sensitivity takes effect. The validity of these means of measure for SoS problems is at present only hypothesized, awaiting verification via simulation as SoS studies ensue.



**Figure 3. Development of Proper SoS Performance Measures with Uncertainty**

### III. Transportation as a System-of-Systems

#### A. Introduction- Transportation is a SoS Problem

There is a pressing need to quantitatively investigate potential transportation concepts on a national scale, i.e., the National Transportation System (NTS), to track the cascade of interrelated technological, infrastructure-related and societal perturbations over significant time horizons. *Modeling and evaluating the behavior of the NTS is overwhelmingly difficult due especially to: the extreme number and heterogeneity of independent systems, the distributed nature of these systems, and the presence of deep uncertainty in understanding their co-evolution.* In light of these general properties, the NTS may best be conceived as a system-of-systems—a collection of diverse things that evolve over time, organized at multiple levels, to achieve a range of (likely) conflicting objectives, but never quite behaving as planned. Moreover, examination of these NTS traits clearly indicates the presence of a preponderance of the SoS traits presented in Table 2. The mapping of generic SoS traits to transportation are offered in Table 4.

**Table 4. Mapping of SoS Traits to the NTS**

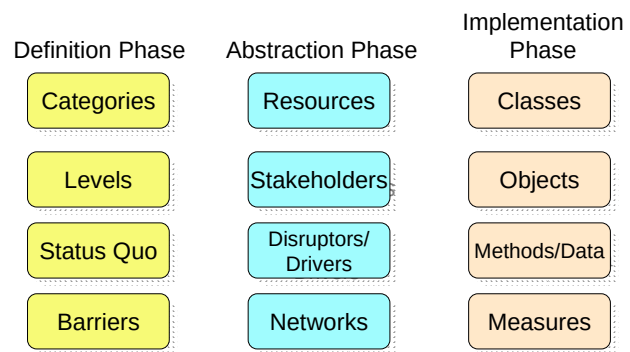
| SoS Trait                             | Transportation Mapping                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operational & Managerial Independence | Numerous transportation systems are operated independent of other systems, though at higher levels some cooperation is evident; e.g. a commercial aircraft operates on a schedule dictated solely by its company, but must coordinate while in flight with other aircraft (traffic). The various transportation business enterprises operate in a similar independent-coordinated vein. |
| Geographic Distribution               | The nature of transportation systems makes obvious the presence of this trait, though the increased sharing of <i>information</i> between distributed systems is of particular interest for the future NTS.                                                                                                                                                                             |
| Evolutionary Behavior                 | Measures of effectiveness for transportation are dynamic in nature, due primarily to delayed response to major inputs as well as inherent feedback mechanisms; static performance measures are not often used (though they are often used for evaluating new systems within the NTS).                                                                                                   |
| Emergent Behavior                     | Some behavior is only evident when two or more effects are combined (e.g. weather in Atlanta during peak evening activity).                                                                                                                                                                                                                                                             |
| Heterogeneity                         | Transportation systems are diverse, including categories such as vehicles (aircraft, trains, autos, etc.), infrastructure, service providers, users, and regulatory bodies.                                                                                                                                                                                                             |
| Networks                              | Transportation network topologies exist for both vehicles and the business enterprises that operate them; each topology can range from pure hub-and-spoke to fully distributed, with hybrid (and possible adaptive) ones also present.                                                                                                                                                  |
| Trans-domain                          | The variety of engineered systems and human organizations (stakeholders) in the transportation SoS indicate the numerous domains that must be spanned for a holistic study approach to be fully adopted.                                                                                                                                                                                |

## B. Context- A “Proto-method” for System-of-Systems

The casting of future transportation concepts as a SoS problem means that the implications of the associated SoS traits discussed in section II must be faced. Therefore, the challenge to provide effective design methods for decision-support in this context must also be faced. Strategies are under development to deal with these challenges and some portions have already been applied to transportation.<sup>16,17</sup> Ongoing research by the author and collaborators have begun to consolidate these strategies into an initial SoS approach- what may be termed a “proto-method”. In the remainder of this paper, a broad outline of this proto-method is presented along with some interim demonstration from application in the transportation domain and reference to articles that contained more detailed results. Undoubtedly, very much remains to be done in this endeavor.

The graphic in Figure 4 gives an overarching view of the major steps in the proto-method for SoS problems as currently conceived. The major activity in the *Definition Phase* is one of understanding- characterize the SoS as it currently exists and establishing categories and levels that will later be required to detect evolutionary and emergent properties. In the *Abstraction Phase*, the main actors, effectors, disturbances, and networks are drawn out and placed in context corresponding to their real inter-relation. The focus of this phase is in grappling with the overwhelming complexity by abstracting the main entities; it is not intended to construct a detailed hierarchical decomposition. Finally, the *Implementation Phase* instantiates all or part of the abstraction within a modeling and simulation environment. It is in this final phase that specific hypotheses about the SoS can be proposed and tested.

The development of the approach summarized in Figure 4 actually occurred in reverse from the phase order displayed. The publication record would prove this true for sure. A modeling approach for dynamic system analyses of new air vehicles was developed first.<sup>16</sup> Next, the abstraction model was established following the recognition of the need for a more holistic perspective for design for future transportation architectures.<sup>17</sup> Finally, the importance of SoS problem definition (esp. categories and levels) prior to abstraction was recognized.<sup>1</sup> Perhaps, the salient lesson in the order of development is this- to better comprehend the complexity of the SoS in abstract, one must have first struggled with “ $\alpha$ -level issues” initially. *More precisely, by learning*

**Figure 4. “Proto-Method” for SoS Problems**

to develop insights on smaller, more manageable levels, a capability and awareness of the larger issues is more likely to emerge (if one is looking).

A final, general note concerning the method is a philosophical one. Analogous to the Internet in many ways, it seems clear that there is no one individual who controls the NTS or its future design. Yet, the questions of interest about a transportation SoS could only completely be answered by such an individual- and “imagined architect”. This hypothetical person would be able to shape an architecture for the NTS by design and would surely realize that not only physical factors, such as vehicles and infrastructure, but also organizational elements, such as public interest groups and industrial firms, should reside inside the problem boundary. These and other notions associated with the imagined architect will appear along the journey of exploring this “proto-method” for transportation.

### C. Definition Phase

The objectives of this phase are to sketch the mental map of the SoS as it exists (embodied in categories, levels, and forces at play) and identify barriers to more preferred behavior. Execution of a portion of this phase for transportation is summarized in Table 5, illustrating how the NTS could be considered in a SoS frame of reference. This use of the lexicon introduced in Table 3 provides value at two levels: first, the breadth of the problem and subsequent imperative to move beyond (across) domain stovepipes is evident, and second, the categorizations help effectively guide the *Abstraction Phase*. The variety of decision-makers involved in transportation can be identified, engaged, and included in the discussion. Through subsequent modeling, the probabilities for solutions at the  $\gamma$ - or  $\delta$ -levels can be formed by aggregating the  $\alpha$ - and  $\beta$ -level entities. It is also important to note the number of entities at each level may vary tremendously, likely by orders of magnitude. For example, in Table 5, estimates are given for the number of entities at each level, ranging from  $10^6$  to  $10^2$  in just two level shifts. The motivation behind the formation of this category/level description can be found in Ref. [1].

**Table 5. Transportation SoS- Lexicon Matrix with Order Estimates**

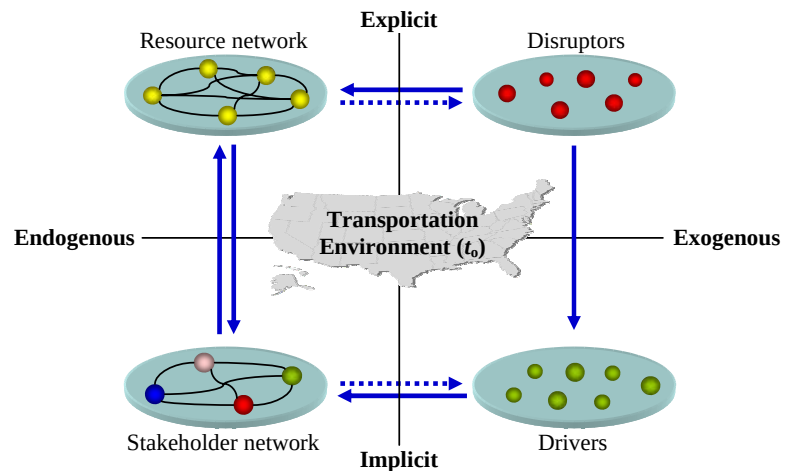
| Level                            | Resources                                                                     | Operations                                                                                       | Economics                                                                        | Policy                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| $\alpha$<br>( $9 \cdot 10^6$ )   | Vehicles & Infrastructure<br>(e.g. aircraft, truck, runway )                  | Operating a Resource<br>(Aircraft, truck, etc.)                                                  | Economics of building/operating/buying/selling /leasing a single resource        | Policies relating to single resource use (e.g. type certification, flight procedures, etc.)  |
| $\beta$<br>( $9 \cdot 10^4$ )    | Collection of resources for a common function<br>(an airport, etc)            | Operating resource networks for common function (e.g. airline)                                   | Economics of operating/buying/selling /leasing resource networks                 | Policies relating to multiple vehicle use (e.g. airport traffic mangt, noise policies, etc.) |
| $\gamma$<br>( $9 \cdot 10^2$ )   | Resources in a Transport Sector (e.g. air transportation)                     | Operating collection of resource networks (e.g. ; commercial air Ops)                            | Economics of a Business sector (e.g. Airline Industry)                           | Policies relating to sectors using multiple vehicles. (safety, accessibility, etc.)          |
| $\delta$<br>( $9 \cdot 10^1$ )   | Multiple, interwoven sectors (resources for a national transportation system) | Operations of Multiple Business Sectors (i.e. Operators of total national transportation system) | Economics of total national transportation system (All Transportation Companies) | Policies relating national transportation policy                                             |
| $\epsilon$<br>( $9 \cdot 10^0$ ) | Global transportation system                                                  | Global Operations in the world transportation system                                             | Global Economics of the world transportation system                              | Policies relating to the global transportation system                                        |

### D. Abstraction Phase

An abstraction for transportation that builds from the levels and categories has been carefully developed and reported.<sup>18</sup> A brief summary is given here to highlight the role of this phase in the SoS proto-method. Two pairs of entity descriptors emerge from the abstraction process: explicit-implicit and endogenous-exogenous. Unlike under the reductionism mindset, the role of the descriptors is not to facilitate break-down of the entities into separate pieces. Instead, it is only to organize them by articulating their inherent natures. Four entity categories are generated based on the descriptors: resources, stakeholders, drivers, and disruptors. All these entities are inter-webbed by networks that define the linkages amongst themselves. This is summarized in Figure 5.

Vehicles and infrastructure are examples of resources that consumers physically experience (explicit) when traveling or sending shipments. Further, they are under partial or full control of the imagined architect, thus endogenous. But there are ‘other-than-physical’ entities that desire to exert forces on the architecture for their own interests. This type of endogenous entity is called stakeholder, and in most circumstances their behaviors and

decisions are not manifested in an explicit manner to the consumers (implicit). The stakeholders reside in both private and public sectors, ranging from the actual consumers of transportation services to the providers of those services. Each stakeholder holds objectives they wish to pursue in the transportation environment. For example, an individual values doorstep-destination (D-D) speed, cost per mph, mobility flexibility, etc. However, from a societal perspective, there may be desires to minimize total energy expended, maximize the robustness of the system to disturbance, etc. Often, members of these two groups move in tandem (e.g. robustness of system and mobility freedom), but other times they may not (e.g. possible increase in energy per mile traveled in an on-demand system).



**Figure 5. Entity-Centric Abstraction Model**

While the stakeholders and resources are considered endogenous building blocks, the transportation environment contains exogenous entities that are outside of the architect's controllable domain. This entity class has been traditionally treated as given assumptions, circumstances, and constraints about the transportation environment (e.g., population, weather). Driver entities are largely concerned with economic, societal, and psychological circumstances that influence the stakeholder network by implicit means. On the other hand, disruptor entities explicitly affect the resource network and/or a portion of the driver entities by reducing the efficiency of the resource network, disabling particular nodes or links of the network.

The network concept has appeared both in the *Definition Phase* and the *Abstraction Phase*. As alluded to in Table 2, networks ("graphs" in the mathematics community) can be viewed at a simple level as constructs that define the connectivity (links) between entities (nodes). The transportation abstraction in Figure 5 identifies networks existing in both the resource and the stakeholder domains. For example, air transportation resource networks arise when nodes (airports) are linked by aircraft either in a scheduled or un-scheduled (on-demand) manner. The connectivity provided by the commercial airline industry is determined by pre-defined schedules and is presently by far the predominant means of air transportation. Charter, fractional ownership, and personally owned options exist, but they remain at least one order of magnitude higher in cost than the scheduled service. Demand, which theoretically drives schedules, is an important component of the real networks. Thus, in contrast to the present network, unscheduled (on-demand) models are also being increasingly studied. On-demand implies that the systems organize themselves to serve a particular demand of a particular customer at a particular time. In this case, ad-hoc networks would be formed to respond to demand, requiring a flexible set of vehicles and traffic management rules to do the job. A very insightful encapsulation of these possible networks as well as the various layers within the resource networks has been presented by Holmes.<sup>19</sup>

But how these networks actually arise, how they evolve, and how robust they are to disruption are topics of current research. Further, such networks can only arise if there is an economic imperative to do so. For example, Crossley et al. studied the problem of a simplified airline as a system-of-systems, employing a combination of MDO techniques with integer programming to determine how a new aircraft design can be optimized for the greater good of the existing airline operation.<sup>20</sup> In another line of work, the author and collaborators have explored a UAV-based logistics network.<sup>21</sup> In that study, comparative studies of SoS performance versus alternative network topologies and vehicle performance levels were generated. The measures of merit were profitability of a service provider attempting to operate this system in a competitive commercial environment. Overall, the entire abstraction phase, including entities and networks, is important for framing the relevant hypotheses about the SoS problem at hand and thus the most appropriate means for implementation.

## E. Implementation Phase

Prospects for a purely analytical solution to most system-of-systems problems are dim at best. Thus, the research direction of most interest for the *Implementation Phase* involves instantiating the aforementioned conceptual abstraction for transportation via simulation. Several elements of such a simulation for this endeavor have already

been created and exercised. However, a more comprehensive simulation is required, one in which the higher levels of organization (the layered networks) can be considered when forming the concept space for future transportation architectures. The simulation must properly characterize the salient SoS traits, including the emergent properties that can arise when individual systems are empowered to adapt and form tailored networks. Further, it is intended to use the extended simulation to draw out recurring patterns within alternate configurations that seem to perform well over an ensemble of plausible scenarios (i.e. they are scalable). Two major topics are outlined with regards to modeling and simulation: alternative system views/methods and schemes for effective implementation.

### 1. System Views and Methods

A recent article by Rouse postulated that, among the implications brought by complex systems<sup>‡</sup>, an important one is to consider the variety of possible “system views”, especially since such views are tightly tied with subsequent modeling and simulation approaches.<sup>22</sup> The four views proposed are: Systems of Hierarchical Mappings, Systems of Uncertain State Equations, Systems of Discontinuous Nonlinear Models, and Systems of Autonomous Agents. Some related methods for each view as well as comments on the applicability to transportation SoS studies are offered in Table 6. While each view (and method) is distinct, the magnitude of most SoS problems may require an integration of several or all of these. A means to that end is discussed in the next section.

**Table 6. System Views, Methods, and Transportation SoS Modeling Implications**

| System View <sup>22</sup>            | Methods                                         | Transportation Modeling Implication                                                                                                                                                                                                                                                                                           |
|--------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Systems of Hierarchical Mapping      | Systems Engineering (SE), Decomposition Methods | The proven record of SE for individual system development demands its consideration for SoS problems, especially techniques for Systems Analysis; However, the shifting nature of transportation systems and networks may limit use of this view in its traditional form.                                                     |
| Systems of Uncertain State Equations | System Dynamics, Control Theory                 | Robustness methods when facing uncertainty are well-established in this View and needed for exploring future transportation SoS; However, the inability to characterize the “plant dynamics” and to handle variables that switch between controllable and uncontrollable may severely limit direct application of algorithms. |
| Systems of Nonlinear Models          | Chaos Theory, Networks                          | The notion that simple mechanisms can be the source of complex behavior in several applications has proved intriguing; Identifying those mechanisms in transportation may be difficult, although the network theory experience may be a start.                                                                                |
| Systems of Autonomous Agents         | Agent-Based Modeling                            | Of all the methods, this one most closely matches the SoS traits for transportation outlined in Table 4 and already has some applications reported.                                                                                                                                                                           |

Two of the methods that are perhaps most distinct from those used in the aerospace vehicle design community are system dynamics and agent-based modeling. They are highlighted here because they both address the implications of SoS problems that are poorly addressed by existing aerospace design approaches: evolutionary and emergent behavior.

The modern field of system dynamics was pioneered in the early 1960’s, and has since matured into an active field often used in system studies and policy analysis.<sup>23,24</sup> Through the use of causal loop diagrams and stock and flow dynamic models that capture key feedback mechanisms, system dynamic approaches attempt to determine policies and functions that result in optimal outcomes over time. In recent years, researchers have focused specifically on business dynamics, emphasizing the identification and improvement of behavior patterns. This field could be particularly applicable to analysis of future transportation service provider business models as they evolve along with the technology and regulatory aspects.<sup>25</sup> The type of time responses notionally shown in Figure 3 above is not uncommon in the system dynamics approach. Potential drawbacks, however, are also apparent, such as the inability to explore emergent behavior as it is presently defined.

Agent-Based Modeling (ABM) is an area growing in diversity of formulation and application. The premise of ABM is that the global behavior of a complex system is derived from the low-level interactions among its constituent elements, whose behavior is encoded based on a variety of factors. Autonomous agents representing these elements are individually defined, as is the environment within which they interact. The collective action of

<sup>‡</sup> We shall, for now, overlook the differences, both semantic and real, between ‘complex systems’ and ‘system-of-systems’.

the agents produces an “emergent behavior” which often produces real world insight, even if the model itself is relatively simple in form. The major strength of ABM comes from the fact that it is a simple and versatile method that is well suited for studies of complex non-linear systems. Whereas agent-based models can be made arbitrarily realistic by capturing processes or mechanisms that drive individual components, they can also be made quite abstract in an attempt to understand the essence of the problem.<sup>26</sup> Agent-based simulations can reveal both qualitative and quantitative properties of the real system, so ABM/S can be deemed as *computational laboratories*<sup>27</sup> to perform experiments to test nearly any kind of imaginable hypotheses. This modeling approach holds significant potential to capture the complex interactions existing among the NTS stakeholders from airlines and air traffic control, to passengers and policy makers.

A significant challenge to the acceptance of ABM simulation results (and, more broadly, results from any of the modeling approaches) is validation. In the sciences, validation comes from the comparison of theory with experiment. In a SoS context, these two are intertwined: the theory and the experiment both are embedded in the simulation for the “true” experimental apparatus is the transportation system itself. Thus, the equivalents of the theoretician and experimental communities in SoS applications will likely continue the struggle to find agreeable means for validating the soundness of the methods, not the preciseness of any prediction.

## 2. Object-Oriented Implementation Schemes

The second topic of the *Implementation Phase* to be discussed concerns the implementation scheme associated with instantiating the collection of networks and entity models into a simulation architecture that is effective, flexible, comprehensive, and verifiable. The chosen system views, associated methods, and data from the  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ -levels must be combined to test a myriad of possible hypotheses, especially at the upper levels of organization. For this task, and object-oriented (OO) approach appears well suited. As the name implies, the OO approach is based on the concept of object. Objects are constructs that are responsible for themselves. When a specific problem is at hand, objects are specific methods that can be called by other objects. When a particular computer program that represents the problem is run, objects exercise their methods with their data, all self-contained. Other terminology from the OO world (classes, patterns, polymorphism, etc.) is also becoming well-known.

The claim to fame for the OO approach is *extensibility*- the ability for the software to adapt easily and quickly to changing requirements. Thus, it is the proposition of the author that an OO implementation paradigm is a necessity to achieve the desired objectives for a SoS problem. Perhaps this is no surprise, as the OO approach is now standard across a wide swath of domains. However, there remain “corners” where this shift is still in its infancy (e.g. aircraft design). Further, there appears to be an intriguingly tight connection in philosophy between the system-of-systems perspective and the OO paradigm, one that is just beginning to be explored (see Table 7). The central role of abstraction in properly building a mental model of the problem is at the heart of this connection. More specifically, though the close connection between the concepts of object, entity (introduced in the *Abstraction Phase*), and agent (introduced earlier in the present *Implementation Phase*) are abundantly clear. The hope is that the use of the generalized and abstract objects with certain attributes and methods that modify those attributes can generate the holistic views required for system-of-systems. Further, OO constructs appear to fulfill the need for rapid exchange of model entities and networks within a transportation system analysis environment. What remains is to identify the obstacles that do exist in OO implementations of transportation SoS simulations. The answers remain to be uncovered.

**Table 7- Philosophical Relation between SoS Traits and OO Principles**

| SoS Trait                             | Object-Oriented Construct/Trait                                                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operational & managerial independence | <i>Objects</i> are imbued with responsibility for their own independent behavior (the implementation form of an “entity”, which is a <i>class</i> for individual systems in a SoS).         |
| Distribution in Networks              | <i>Objects</i> are not just resources, but also can represent distributed behavior of other objects.                                                                                        |
| Emergent behavior                     | <i>Polymorphism</i> : The same calls to different derivations of the same class produce different results; Supports the notion that the same entities appear differently in separate views. |
| Heterogeneity*                        | Separate <i>abstract class</i> represents each family of conceptually related systems; <i>Concrete class</i> defines the methods and data types for different systems.                      |
| Trans-domain*                         | The <i>extensibility</i> claim-to-fame for OO approaches indicates hope for wide applicability across domains through provision of interfaces.                                              |

## IV. Summary

The emerging problem class termed system-of-systems is introduced in this paper in a context that relates to aerospace systems design. The premise is that aerospace systems are increasingly seen as a mere part of larger system-of-systems endeavors, and thus design of aerospace systems must be understood in this light. An initial survey of the traits of system-of-systems and the mapping of these traits to an application domain of interest – transportation – indicate that indeed there is urgent need for new methods. Towards this end, a summary of a “proto-method” that has been developed is presented, including discussions of its three major phases: Definition, Abstraction, and Implementation. Within each, steps are taken to ensure that the most challenging behaviors found in systems-of-systems are properly treated: evolutionary and emergent behavior (which result from independent systems operating together in networks). This grander web exhibits new traits not observed in the collection of systems viewed in isolation. It is these hard-to-observe traits that are the real goal of design in this context. This emergent behavior and the already present complication of evolutionary behavior that proceeds over several time scales (some even generational) summarizes perhaps the greatest challenge posed by the SoS problem class.

While the ultimate fruits of a system-of-systems field of study are still a ways off, even conceiving of a larger frame of reference provides increased understanding for our seemingly complex pursuit of better aerospace vehicles. In this larger frame, the ill-advised practice of individual system (e.g. aircraft, air traffic rule, infrastructure, etc) optimization may become more clear as the role of other systems and their connectivity becomes evident. By acknowledging the presence of the higher levels of organization, we can work out some of the vexing challenges facing our studies and our designs for tomorrow’s aerospace creations. The success of future aerospace vehicles will depend on how well they harmonize with the dynamics of the system-of-systems of which they may have membership. The success of future systems-of-systems will depend on the extent to which design methods can properly pose and solve problems in this context.

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